

Realimentación de colector

RE sin desacoplar (sin condensador de emisor)

Datos: $V_{CC}=10V$ · $R_F=250k\Omega$ · $R_C=4.7k\Omega$ · $R_E=1.2k\Omega$ · $\beta=90$ ·
 $C_{in}=10\mu F$ · $C_{out}=10\mu F$

DC:

$$I_B = (V_{CC} - V_{BE}) / (R_F + \beta \cdot (R_C + R_E)) = 11.9 \mu A$$

$$I_C = \beta \cdot I_B = 1.07 \text{ mA}$$

$$I_E = (\beta + 1) \cdot I_B = 1.08 \text{ mA}$$

$$r_e = 26 \text{ mV} / I_E = 24 \Omega$$

$$V_{CE} = V_{CC} - I_C \cdot R_C - I_E \cdot R_E = 3.66 \text{ V}$$

AC:

$$A_v = -R_C / (r_e + R_E) = -3.84$$

$$Z_i = \beta(r_e + R_E) \parallel R_F / (1 - A_v) = 35.2 \text{ k}\Omega$$

$$Z_o = R_C \parallel R_F = 4.61 \text{ k}\Omega$$

$$\text{recordatorio: } r_e = 26 \text{ mV} / I_E$$

Divisor de voltaje

RE desacoplado ($C_E = 50 \mu F$)

Datos: $V_{CC}=22V \cdot R_1=39k\Omega \cdot R_2=3.9k\Omega \cdot R_C=10k\Omega \cdot R_E=1.5k\Omega \cdot \beta=100$
 $\cdot C_{in}=10\mu F \cdot C_{out}=10\mu F \cdot C_E=50\mu F$

DC:

$$V_B = V_{CC} \cdot R_2 / (R_1 + R_2) = 2.0 V$$

$$V_E = V_B - V_{BE} = 1.3 V$$

$$I_E = V_E / R_E = 0.867 \text{ mA}$$

$$I_C \approx I_E = 0.867 \text{ mA}$$

$$r_e = 26\text{mV} / I_E = 30 \Omega$$

$$V_{CE} = V_{CC} - I_C \cdot (R_C + R_E) = 12.03 V$$

AC:

$$A_v = -R_C / r_e = -333$$

$$Z_i = R_1 \parallel R_2 \parallel \beta \cdot r_e = 1.63 \text{ k}\Omega$$

$$Z_o = R_C = 10 \text{ k}\Omega$$

$$\text{recordatorio: } r_e = 26\text{mV} / I_E$$

Polarización de emisor

RB única + RE desacoplado ($C_E = 40 \mu F$)

Datos: $V_{CC}=20V \cdot R_B=430k\Omega \cdot R_C=2k\Omega \cdot R_E=1k\Omega \cdot \beta=50 \cdot$
 $C_{in}=10\mu F \cdot C_{out}=10\mu F \cdot C_E=40\mu F$

DC:

$$I_B = (V_{CC} - V_{BE}) / (R_B + (\beta+1) \cdot R_E) = 40.1 \mu A$$

$$I_C = \beta \cdot I_B = 2.01 \text{ mA}$$

$$I_E = (\beta+1) \cdot I_B = 2.05 \text{ mA}$$

$$r_e = 26\text{mV} / I_E = 12.7 \Omega$$

$$V_{CE} = V_{CC} - I_C \cdot R_C - I_E \cdot R_E = 13.94 \text{ V}$$

AC:

$$A_v = -R_C / r_e = -157$$

$$Z_i = R_B \parallel \beta \cdot r_e = 635 \Omega$$

$$Z_o = R_C = 2 \text{ k}\Omega$$

$$\text{recordatorio: } r_e = 26\text{mV} / I_E$$